Chapter 4 : Principles of Inheritance and Variation

1. What is the pattern of inheritance for polygenic trait? **[NEET-2025]**

   (1) Mendelian inheritance pattern

   (2) Non-mendelian inheritance pattern

   (3) Autosomal dominant pattern

   (4) X-linked recessive inheritance pattern

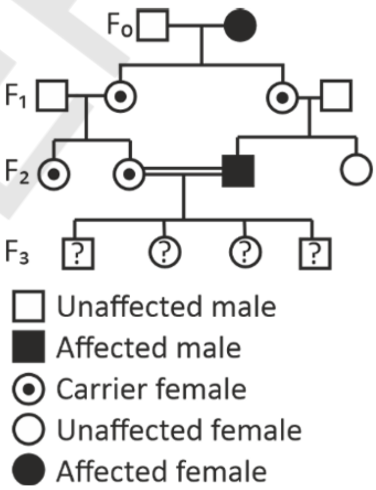
2. Genes R and Y follow independent assortment. If RRYY produce round yellow seeds and rryy produce wrinkled green seeds, what will be the phenotypic ratio of the F2 generation?

   (1) Phenotypic ratio - 1 : 2 : 1 (2) Phenotypic ratio - 3 : 1 **[NEET-2025]**

   (3) Phenotypic ratio - 9 : 3 : 3 : 1 (4) Phenotypic ratio - 9 : 7

3. With the help of given pedigree, find out the probability for the birth of a child having no disease and being a carrier (has the disease mutation in one allele of the gene) in F3 generation.

[NEET-2025]



   (1) 1/4 (2) 1/2

   (3) 1/8 (4) Zero

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4. Given below are two statements: **[Re-NEET-2024]**

Statement I: When many alleles of a single gene govern a character, it is called polygenic inheritance.

Statement II: In Polygenic inheritance, the effect of each allele is additive.

In the light of the above statements, choose the correct answer from the options given below.

(1) Statement I is true but Statement II is false

(2) Statement I is false but Statement II is true

(3) Both Statement I and Statement II are true

(4) Both Statement I and Statement II are false

5. Match List-I with List-II: **[Re-NEET-2024]**

List-I	List-II
A. Incomplete dominance	I. Blood groups in human
B. Co-dominance	II. Flower colour in Antirrhinum
C. Pleiotropy	III. Skin colour in human
D. Polygenic inheritance	IV. Phenylketonuria

Choose the correct answer from the options given below:

(1) A-III, B-IV, C-II, D-I

(2) A-II, B-I, C-IV, D-III

(3) A-II, B-III, C-I, D-IV

(4) A-IV, B-I, C-III, D-II

6. Given below are two statements: **[Re-NEET-2024]**

Statement I: Failure of segregation of chromatids during cell cycle resulting in the gain or loss of whole set of chromosome in an organism is known as aneuploidy.

Statement II: Failure of cytokinesis after anaphase stage of cell division results in the gain or loss of a chromosome is called polyploidy.

In the light of the above statements, choose the correct answer from the options given below:

(1) Statement I is true but Statement II is false

(2) Statement I is false but Statement II is true

(3) Both Statement I and Statement II are true

(4) Both Statement I and Statement II are false

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7. When a tall pea plant with round seeds was selfed, it produced the progeny of : **[Re-NEET-2024]**

(a) Tall plants with round seeds and

(b) Tall plants with wrinkled seeds.

Identify the genotype of the parent plant.

(1) TtRr (2) TtRR

(3) TTRR (4) TTRr

8. Aneuploidy is a chromosomal disorder where chromosome number is not the exact copy of its haploid set of chromosomes, due to : **[Re-NEET-2024]**

A. Substitution B. Addition

C. Deletion D. Translocation

E. Inversion

Choose the most appropriate answer from the options given below :

(1) C and D only (2) D and E only

(3) A and B only (4) B and C only

9. Match List I with List II **[NEET-2024]**

List I	List II
A. Two or more alternative forms of a gene	I. Back cross
B. Cross of F1 progeny with homozygous recessive parent	II. Ploidy
C. Cross of F1 progeny with any of the parents	III. Allele
D. Number of chromosome sets in plant	IV. Test cross

Choose the correct answer from the options given below:

(1) A-I, B-II, C-III, D-IV (2) A-II, B-I, C-III, D-IV

(3) A-III, B-IV, C-I, D-II (4) A-IV, B-III, C-II, D-I

10. Which one of the following can be explained on the basis of Mendel's Law of Dominance?

A. Out of one pair of factors one is dominant and the other is recessive. **[NEET-2024]**

B. Alleles do not show any expression and both the characters appear as such in F2 generation.

C. Factors occur in pairs in normal diploid plants.

D. The discrete unit controlling a particular character is called factor.

E. The expression of only one of the parental characters is found in a monohybrid cross.

Choose the correct answer from the options given below:

(1) A, B and C only

(2) A, C, D and E only

(3) B, C and D only

(4) A, B, C, D and E

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11. A pink flowered Snapdragon plant was crossed with a red flowered Snapdragon plant. What type of phenotype/s is/are expected in the progeny? **[NEET-2024]**

(1) Only red flowered plants

(2) Red flowered as well as pink flowered plants

(3) Only pink flowered plants

(4) Red, Pink as well as white flowered plants

12. In a plant, black seed color (BB/Bb) is dominant over white seed color (bb). In order to find out the genotype of the black seed plant, with which of the following genotype will you cross it?

(1) BB (2) bb **[NEET-2024]**

(3) Bb (4) BB/Bb

13. Match List I with List II : **[NEET-2024]**

List I	List II
A. Down's syndrome	I. 11th chromosome
B. α-Thalassemia	II. 'X' chromosome
C. β-Thalassemia	III. 21st chromosome
D. Klinefelter's syndrome	IV. 16th chromosome

(1) A-I, B-II, C-III, D-IV (2) A-II, B-III, C-IV, D-I

(3) A-III, B-IV, C-I, D-II (4) A-IV, B-I, C-II, D-III

14. Match List-I with List-II. **[NEET-2023-MANIPUR]**

List-I	List-II
(A) Monohybrid Cross	(I) 1 : 1
(B) Dihybrid Cross	(II) 1 : 2 : 1
(C) Incomplete dominance	(III) 3 : 1
(D) Test Cross	(IV) 9 : 3 : 3 : 1

Choose the correct answer from the options given below :

(1) (A)-(II), (B)-(III), (C)-(IV), (D)-(I) (2) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

(3) (A)-(III), (B)-(IV), (C)-(II), (D)-(I) (4) (A)-(II), (B)-(IV), (C)-(III), (D)-(I)

15. A heterozygous pea plant with violet flowers was crossed with a homozygous pea plant with white flowers. Violet is dominant over white. Which one of the following represents the expected combinations among 40 progenies formed? **[NEET-2023-MANIPUR]**

(1) All 40 produced violet flowers

(2) All 40 produced white flowers

(3) 30 produced violet and 10 produced white flowers

(4) 20 produced violet and 20 produced white flowers

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16. In which disorder change of single base pair in the gene for beta globin chain results in change of glutamic acid to valine? **[NEET-2023-MANIPUR]**

(1) Haemophilia

(2) Phenylketonuria

(3) Thalassemia

(4) Sickle cell anaemia

17. A certain plant homozygous for yellow seeds and red flowers was crossed with a plant homozygous for green seeds and white flowers. The F1 plants had yellow seeds and pink flowers. The F1 plants were selfed to get F2 progeny. Assuming independent assortment of the two characters, how many phenotypic categories are expected for these characters in the F2 generation?

(1) 4 (2) 6

(3) 9 (4) 16

18. Select the correct statements about sickle cell anaemia. **[NEET-2023-MANIPUR]**

(A) There is a change in gene for beta globin.

(B) In the beta globin, there is valine in the place of Lysine.

(C) It is an example of point mutation.

(D) In the normal gene U is replaced by A.

Choose the correct answer from the options given below:

(1) (A), (B) and (D) only (2) (A) and (C) only

(3) (B), (C) and (D) only (4) (B) and (D) only

19. Select the correct statement/s with respect to mechanism of sex determination in Grasshopper.

(A) It is an example of female heterogamety. **[NEET-2023-MANIPUR]**

(B) Male produces two different types of gametes either with or without X chromosome.

(C) Total number of chromosomes (autosomes and sex chromosomes) is same in both males and females.

(D) All eggs bear an additional X chromosome besides the autosomes.

Choose the correct answer from the options given below :

(1) (A) only

(2) (A) and (C) only

(3) (B) and (D) only

(4) (A), (C) and (D) only

20. Broad palm with single palm crease is visible in a person suffering from **[NEET-2023]**

(1) Down's syndrome (2) Turner's Syndrome

(3) Klinefelter Syndrome (4) Thalassemia

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21. Frequency of recombination between gene pairs on same chromosome as a measure of the distance between genes to map their position on chromosome, was used for the first time by **[NEET-2023]**

(1) Thomas Hunt Morgan (2) Sutton and Boveri

(3) Alfred Sturtevant (4) Henking

22. The phenomenon of pleiotropism refers to **[NEET-2023]**

(1) Presence of several alleles of a single gene controlling a single crossover

(2) Presence of two alleles, each of the two genes controlling a single trait

(3) A single gene affecting multiple phenotypic expression

(4) More than two genes affecting a single character

23. Which of the following statements are correct about Klinefelter's Syndrome? **[NEET-2023]**

A. This disorder was first described by Langdon Down (1866).

B. Such an individual has overall masculine development. However, the feminine development is also expressed.

C. The affected individual is short statured.

D. Physical, psychomotor and mental development is retarded.

E. Such individuals are sterile.

Choose the correct answer from the options given below:

(1) A and B only (2) C and D only

(3) B and E only (4) A and E only

24. Which one of the following symbols represents mating between relatives in human pedigree analysis? **[NEET-2023]**

25. Given below are two statements: **[NEET-2022]**

Statement I: Mendel studied seven pairs of contrasting traits in pea plants and proposed the Laws of Inheritance

Statement II: Seven characters examined by Mendel in his experiment on pea plants were seed shape and colour, flower colour, pod shape and colour, flower position and stem height

In the light of the above statements, choose the correct answer from the options given below :-

(1) Both Statement I and Statement II are incorrect

(2) Statement I is correct but Statement II is incorrect

(3) Statement I is incorrect but Statement II is correct

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(4) Both Statement I and Statement II are correct

26. XO type of sex determination can be found in:- **[NEET-2022]**

(1) Birds (2) Grasshoppers

(3) Monkeys (4) Drosophila

27. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R). **[NEET-2022]**

Assertion (A): Mendel's law of Independent assortment does not hold good for the genes that are located closely on the same chromosome.

Reason (R): Closely located genes assort independently.

In the light of the above statements, choose the correct answer from the options given below:

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(2) (A) is correct but (R) is not correct

(3) (A) is not correct but (R) is correct

(4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

28. Which of the following occurs due to the presence of autosome linked dominant trait ?

(1) Myotonic dystrophy (2) Haemophilia **[NEET-2022]**

(3) Thalassemia (4) Sickle cell anaemia

29. The recombination frequency between the genes a & c is 5%, b & c is 15%, b & d is 9%, a & b is 20%, c & d is 24% and a & d is 29%. What will be the sequence of these genes on a linear chromosome ? **[NEET-2022]**

(1) d, b, a, c (2) a, b, c, d

(3) a, c, b, d (4) a, d, b, c

30. If a colour blind female marries a man whose mother was also colour blind, what are the chances of her progeny having colour blindness ? **[NEET-2022]**

(1) 50% (2) 75%

(3) 100% (4) 25%

31. Given below are two statements :- **[Re-NEET-2022]**

Statement I : Sickle cell anaemia and Haemophilia are autosomal dominant traits.

Statement II : Sickle cell anaemia and Haemophilia are disorders of the blood.

In the light of the above statements, choose the correct answer from the options given below :

(1) Both Statement I and Statement II are correct

(2) Both Statement I and Statement II are incorrect

(3) Statement I is correct but Statement II is incorrect

(4) Statement I is incorrect but Statement II is correct

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32. The chromosomal theory of inheritance was proposed by :- **[Re-NEET-2022]**

(1) Thomas Morgan (2) Sutton and Boveri

(3) Gregor Mendel (4) Robert Brown

33. If a female individual is with small round head, furrowed tongue, partially open mouth and broad palm with characteristic palm crease. Also the physical, psychomotor and mental development is retarded. The karyotype analysis of such an individual will show :- **[Re-NEET-2022]**

(1) 47 chromosomes with XXY sex chromosomes

(2) 45 chromosomes with XO sex chromosomes

(3) 47 chromosomes with XYY sex chromosomes

(4) Trisomy of chromosome 21

34. What is the expected percentage of F2 progeny with yellow and inflated pod in dihybrid cross experiment involving pea plants with green coloured, inflated pod and yellow coloured constricted pod ? **[Re-NEET-2022]**

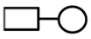
(1) 100% (2) 56.25%

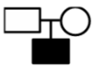
(3) 18.75 % (4) 9%

35. Select the incorrect match regarding the symbols used in Pedigree analysis :- **[Re-NEET-2022]**

(1)  — Sex unspecified

(2)  — Affected individual

(3)  — Consanguineous mating

(4)  — Parent with male child affected with disease

36. A normal girl, whose mother is haemophilic marries a male with no ancestral history of haemophilia. What will be the possible phenotypes of the offsprings ? **[Re-NEET-2022]**

(a) Haemophilic son and haemophilic daughter.

(b) Haemophilic son and carrier daughter.

(c) Normal daughter and normal son.

(d) Normal son and haemophilic daughter.

Choose the most appropriate answer from the options given below :

(1) (a) and (b) only (2) (b) and (c) only

(3) (a) and (d) only (4) (b) and (d) only

37. The production of gametes by the parents, formation of zygotes, the F1 and F2 plants, can be understood from a diagram called :- **[NEET-2021]**

(1) Bullet square (2) Punch square

(3) Punnett square (4) Net square

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38. In a cross between a male and female, both heterozygous for sickle cell anaemia gene, what percentage of the progeny will be diseased ? **[NEET-2021]**

(1) 50% (2) 75%

(3) 25% (4) 100%

39. Experimental verification of the chromosomal theory of inheritance was done by- **[NEET-2020]**

(1) Morgan (2) Mendel

(3) Sutton (4) Boveri

40. Select the correct match :- **[NEET-2020]**

(1) Thalassemia - X linked

(2) Haemophilia - Y linked

(3) Phenylketonuria - Autosomal dominant trait

(4) Sickle cell anaemia - Autosomal recessive trait, chromosome-11

41. Identify the wrong statement with reference to the gene 'I' that controls ABO blood groups:

(1) Allele 'i' does not produce any sugar. **[NEET-2020]**

(2) The gene (I) has three alleles.

(3) A person will have only two of the three alleles.

(4) When IA and IB are present together, they express same type of sugar.

42. How many true breeding pea plant varieties did Mendel select as pairs, which were similar except in one character with contrasting traits? **[NEET-2020]**

(1) 8 (2) 4

(3) 2 (4) 14

43. Select the correct match. **[NEET-2020]**

(1) Thalassemia - X linked

(2) Haemophilia - Y linked

(3) Phenylketonuria - Autosomal dominant trait

(4) Sickle cell anaemia - Autosomal recessive trait, chromosome – 11

44. The number of contrasting characters studied by Mendel for his experiments was :-

(1) 14 (2) 4 **[NEET-2020(COVID-19)]**

(3) 2 (4) 7

45. The best example for pleiotropy is :- **[NEET-2020(COVID-19)]**

(1) Skin colour

(2) Phenylketonuria

(3) Colour Blindness

(4) ABO Blood group

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46. Chromosomal theory of inheritance was proposed by :- **[NEET-2020(COVID-19)]**

(1) Sutton and Boveri

(2) Bateson and Punnet

(3) T. H. Morgan

(4) Watson and Crick

47. What map unit (Centimorgan) is adopted in the construction of genetic maps? **[NEET-2019]**

(1) A unit of distance between two expressed genes, representing 10% cross over

(2) A unit of distance between two expressed genes, representing 100% cross over

(3) A unit of distance between genes on chromosomes, representing 1% cross over

(4) A unit of distance between genes on chromosomes, representing 50% cross over

48. A gene locus has two alleles A, a. If the frequency of dominant allele A is 0.4 then what will be the frequency of homozygous dominant, heterozygous and homozygous recessive individuals in the population ? **[NEET-2019]**

(1) 0.36 (AA); 0.48(Aa); 0.16 (aa) (2) 0.16 (AA); 0.24 (Aa); 0.36 (aa)

(3) 0.16 (AA); 0.48 (Aa); 0.36 (aa) (4) 0.16 (AA); 0.36 (Aa); 0.48 (aa)

49. The frequency of recombination between gene pairs on the same chromosome as a measure of the distance between genes was explained by :- **[NEET-2019]**

(1) T.H. Morgan (2) Gregor J. Mendel

(3) Alfred Sturtevant (4) Sutton Boveri

50. In Antirrhinum (Snapdragon), a red flower was crossed with a white flower and in F1 generation, pink flowers were obtained. When pink flowers were selfed, the F2 generation showed white, red and pink flowers. Choose the incorrect statement from the following :- **[NEET-2019]**

(1) This experiment does not follow the Principle of Dominance

(2) Pink colour in F1 is due to incomplete dominance.

(3) Ratio of F2 is 1/ 4(Red) : 2 / 4(Pink) : 1 / 4(White)

(4) Law of Segregation does not apply in this experiment.

51. Select the incorrect statement. **[NEET-2019]**

(1) Male fruit fly is heterogametic.

(2) In male grasshoppers, 50% of sperms have no sex-chromosome.

(3) In domesticated fowls sex of progeny depends on the type of sperm rather than egg.

(4) Human males have one of their sex chromosome much shorter than the other.

52. In which genetic condition, each cell in the affected person has three sex chromosomes XXY ?

(1) Thalassemia (2) Kleinfelter's Syndrome **[NEET-2019(Odisha)]**

(3) Phenylketonuria (4) Turner's Syndrome

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53. In a marriage between male with blood group A and female with blood group B, the progeny had either blood group AB or B. What could be the possible genotype of parents ?

(1) IAi (Male) : IBIB(Female)	(2) IAIA (Male) : IBIB(Female)	**[NEET-2019(Odisha)]**

(3) IAIA(Male) : IBi (Female)	(4) IAi (Male) : IBi (Female)

54. The production of gametes by the parents, the formation of zygotes, the F1 and F2 plants, can be understood using :- **[NEET-2019(Odisha)]**

(1) Pie diagram	(2) A pyramid diagram

(3) Punnet square	(4) Venn diagram

55. Select the correct statement :- **[NEET-2018]**

(1) Franklin Stahl coined the term "linkage".

(2) Punnett square was developed by a British scientist.

(3) Spliceosomes take part in translation.

(4) Transduction was discovered by S. Altman.

56. Which of the following pairs in wrongly matched ? **[NEET-2018]**

(1) Starch synthesis in pea : Multiple alleles

(2) ABO blood grouping : Co-dominance

(3) XO type sex determination : Grasshopper

(4) T.H. Morgan : Linkage

57. Which of the following characteristics represent 'Inheritance of blood groups' in humans ?

a. Dominance	b. Co-dominance	**[NEET-2018]**

c. Multiple allele	d. Incomplete dominance

e. Polygenic inheritance

(1) b, c and e	(2) a, b and c

(3) b, d and e	(4) a, c and e

58. A woman has an X-linked condition on one of her X chromosomes. This chromosome can be inherited by :- **[NEET-2018]**

(1) Only daughters	(2) Only sons

(3) Only grandchildren	(4) Both sons & daughters

59. Thalassemia and sickle cell anemia are caused due to a problem in globin molecule synthesis. Select the correct statement :- **[NEET-2017]**

(1) Both are due to a quantitative defect in globin chain synthesis

(2) Thalassemia is due to less synthesis of globin molecules

(3) Sickel cell anemia is due to a quantitative problem of globin molecules

(4) Both are due to a qualitative defect in globin chain synthesis

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60. The genotypes of a husband and Wife are IAIB & IA i. Among the blood types of their children, how many different genotypes and phenotypes are possible? **[NEET-2017]**

(1) 3 genotypes ; 4 phenotypes (2) 4 genotypes ; 3 phenotypes

(3) 4 genotypes ; 4 phenotypes (4) 3 genotypes ; 3 phenotypes

61. Which one from those given below is the period for Mendel's hybridization experiments ?

(1) 1840 - 1850 (2) 1857 - 1869 **[NEET-2017]**

(3) 1870 - 1877 (4) 1856 – 1863

62. A disease caused by an autosomal primary non-disjunction is : **[NEET-2017]**

(1) Klinefelter's Syndrome (2) Turner's Syndrome

(3) Sickel Cell Anaemia (4) Down's Syndrome

63. Which of the following most appropriately describes haemophilia ? **[NEET-I 2016]**

(1) Recessive gene disorder (2) X - linked recessive gene disorder

(3) Chromosomal disorder (4) Dominant gene disorder

64. A tall true breeding garden pea plant is crossed with a dwarf true breeding garden pea plant. When the F1 plants were selfed the resulting genotypes were in the ratio of :-

(1) 1 : 2 : 1 :: Tall homozygous : Tall heterozygous : Dwarf **[NEET-I 2016]**

(2) 1 : 2 : 1 :: Tall heterozygous : Tall homozygous : Dwarf

(3) 3 : 1 :: Tall : Dwarf

(4) 3 : 1 :: Dwarf : Tall

65. Pick out the correct statements :- **[NEET-I 2016]**

(a) Haemophilia is a sex-linked recessive disease

(b) Down's syndrome is due to aneuploidy

(c) Phenylketonuria is an autosomal recessive gene disorder.

(d) Sickle cell anaemia is a X-linked recessive gene disorder

(1) (a) and (d) are correct (2) (b) and (d) are correct

(3) (a), (c) and (d) are correct (4) (a), (b) and (c) are correct

66. In a testcross involving F1 dihybrid flies, more parental-type offspring were produced than the recombinant-type offspring. This indicates: **[NEET-I 2016]**

(1) The two genes are located on two different chromosomes.

(2) Chromosomes failed to separate during meiosis.

(3) The two genes are linked and present on the same chromosome.

(4) Both of the characters are controlled by more than one gene.

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67. A true breeding plant is :- **[NEET-II 2016]**

(1) near homozygous and produces offspring of its own kind

(2) always homozygous recessive in its genetic constitution

(3) one that is able to breed on its own

(4) produced due to cross-pollination among unrelated plants

68. If a colour-blind man marries a woman who is homozygous for normal colour vision, the probability of their son being colour-blind is:- **[NEET-II 2016]**

(1) 0.75

(2) 1

(3) 0

(4) 0.5

69. Genetic drift operates in :- **[NEET-II 2016]**

(1) Non-reproductive population

(2) Slow reproductive population

(3) Small isolated population

(4) Large isolated population

70. In Hardy-Weinberg equation, the frequency of heterozygous individual is represented by:-

(1) pq **[NEET-II 2016]**

(2) q2

(3) p2

(4) 2pq

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ANSWER KEY

1. (2)
2. (3)
3. (1)
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70. (4)

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